APPLICATION BASED ASSESSMENT TOOL-1

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Scenario 1: Event Communication Security (Caesar Cipher)

a) Encryption

The original message is:

Plaintext: MEET AT NOON

A Caesar cipher with a **shift of 4** moves every letter four places forward in the alphabet.

Plaintext	**M**	**E**	**E**	**T**	**A**	**T**	**N**	**O**	**O**	**N**
Ciphertext	Q	I	I	X	E	X	R	S	S	R

Encrypted Message: QIIX EX RSSR

b) Decryption

The receiver knows that a shift of 4 was used. By moving each letter **four positions backward**, the original message is recovered.

Decrypted Message: MEET AT NOON

Scenario 2: Intercepted Suspicious Message

Given Ciphertext

GSRH RH Z HVXIVG

Original Message

The message is encrypted using the **Atbash Cipher**, where each letter is replaced with its opposite in the alphabet (A↔Z, B↔Y, C↔X, etc.).

Ciphertext	**GSRH**	**RH**	**Z**	**HVXIVG**
Plaintext	THIS	IS	A	SECRET

Original Message: THIS IS A SECRET

Reasoning

- The short word **RH** converts to **IS**, a common English word.
- The single letter **Z** becomes **A**.
- The final word translates to **SECRET**, forming a meaningful sentence.
- This pattern matches the characteristics of the Atbash substitution cipher.

Scenario 3: Secure Messaging System (Vigenère Cipher)

Plaintext

HELLO

Keyword

KEY

The keyword is repeated until it matches the message length.

Plaintext	**H**	**E**	**L**	**L**	**O**
Keyword	K	E	Y	K	E
Ciphertext	R	I	J	V	S

Encrypted Message: RIJVS

Explanation

- The keyword **KEY** is repeated as **KEYKE**.
- Each letter of the keyword provides a different shift value.
- Since every character is shifted differently, the ciphertext is more secure than a simple Caesar cipher.

Scenario 4: Suspicious Digital Asset Investigation

Detecting Hidden Information

Investigators can examine the image using the following methods:

1. **Least Significant Bit (LSB) Analysis**
    - Checks whether image pixels have been modified to store hidden information.

2. **Steganalysis Tools**
    - Specialized forensic software analyzes the image for abnormal patterns or hidden files.

Methods to Improve Security

1. **Encrypt the secret message before embedding it**
    - Even if the hidden data is extracted, it cannot be understood without the encryption key.
2. **Use Random Embedding**
    - Store the data in randomly selected pixels instead of consecutive pixels, making detection much harder.

Scenario 5: Legacy Encryption in Corporate Communication

a) Original Message

Ciphertext: WECRLTEERDSOEEFEAOCAIVDEN

After reversing the **3-rail Rail Fence Cipher**, the original message is:

WE ARE DISCOVERED FLEE AT ONCE

b) Weaknesses of the Method

- It only rearranges the letters without changing them.
- The number of possible rail patterns is very small.
- Modern computers can quickly try all arrangements and recover the original text.
- It does not provide strong protection for confidential communication.

c) Suggested Improvement

Use a modern encryption algorithm such as **AES (Advanced Encryption Standard)**.

Advantages:

- Uses a secret cryptographic key.
- Provides strong confidentiality.
- Resistant to brute-force and cryptographic attacks.
- Suitable for today's secure communication systems.